

Blockchain-Based Medical Record System

Problem Statement

The Idea

Problem: Health records are insecure and fragmented across multiple systems.

Solution: Store medical records securely using blockchain technology.

Key Features:

- Patient-controlled access permissions
- Tamper-proof storage and verification
- Smart contracts for automated compliance

Current Healthcare Data Challenges

Healthcare data today is fragmented, insecure, and difficult for patients to control. Medical records are typically stored in isolated hospital databases, creating data silos that prevent seamless sharing between providers. This fragmentation leads to several critical problems:

- **Repeated medical tests** due to inability to access previous results
- **Delayed treatments** from incomplete patient histories
- **Increased healthcare costs** from inefficiencies and redundancies

Security and Privacy Risks

Centralized storage systems present significant vulnerabilities:

- **Data breaches:** Single points of failure make entire databases vulnerable to cyberattacks
- **Unauthorized access:** Limited audit trails make it difficult to track who accessed patient data
- **Record tampering:** No cryptographic verification to ensure data integrity
- **Lack of patient control:** Patients have minimal visibility into who accesses their sensitive medical information

Regulatory Compliance Challenges

In regions with strict privacy regulations such as GDPR in Europe, healthcare institutions face complex and costly compliance requirements:

- Ensuring secure storage with proper encryption and access controls

- Maintaining comprehensive auditability of all data access and modifications
- Managing patient consent for data sharing across organizations
- Implementing data deletion mechanisms while preserving compliance records

Essential System Requirements

To address these challenges, a comprehensive solution must ensure:

- **Secure storage of medical records** with end-to-end encryption
- **Patient ownership and access control** enabling individuals to manage their own data
- **Tamper-proof audit trails** providing cryptographic verification of all activities
- **Interoperability across healthcare providers** allowing seamless data exchange
- **Regulatory-compliant data handling** meeting GDPR, HIPAA, and other privacy standards